Student's t and Related Distributions

CSU, Chico Math 351

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Recap: Using σ

Consider a random sample of n measurements, $X_1, \ldots, X_n$, describing a trait believed to be normally distributed. Frequently we want to find the sample mean $\bar{X}$ to estimate the population mean μ.

Confidence intervals and hypothesis tests (up until now) required knowledge of the population standard deviation, σ:

$$\bar{X} \pm z^*_{1-\alpha/2} \cdot \sigma_{\bar{X}}, \quad \textit{and} \qquad z = \frac{\bar{X} - \mu_0}{\sigma_{\bar{X}}}.$$

where $\sigma_{\bar{X}} = \sigma/\sqrt{n}$.

However, σ is frequently unknown, much like the μ we're trying to estimate.

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Confidence Intervals, from now on

Confidence intervals and hypothesis tests (from now on) do not require knowledge of σ. Instead we can use **S**, which is the standard deviation of the sample itself. The variance S^2 is defined as:

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

From this, we can calculate the **sample error**, $S_{\bar{X}} = \frac{S}{\sqrt{n}}$.

Rather than using the Normal distribution, where

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$$

we now use the **Student t distribution**, where

$$t = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1}$$

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Distribution of a Statistic

The Normal, Uniform, Exponential, and other distributions model individual measurements, but some distributions model a statistic. We've seen the Normal distribution can be also used for this (the distribution of $\bar{X}$) thanks to the Central Limit Theorem.
Sampling Distributions tied to the Normal distribution include:

- χ^2 Distribution
- t Distribution (from previous slide)
- F distribution

χ^2 Distribution

Let $Z_1, ..., Z_n$ be n iid standard normal random variables. A random variable with the same distribution as $\sum_{i=1}^{n} Z_i^2$ is a **Chi-squared random variable** with n degrees of freedom.

$$\chi_n^2 = \sum_{i=1}^{n} Z_i^2$$

χ^2 Distribution

This is technically a form of the Gamma distribution, which is:

$$f(x|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, x > 0$$

$$\Gamma(\alpha) = (\alpha - 1)!$$

A Chi-Square random variable with n degrees of freedom will follow a Gamma distribution where $\alpha = n/2$ and $\beta = 1/2$:

$$\sum_{i=1}^{n} Z_i^2 \sim Gamma(n/2, 1/2)$$

χ^2 Distribution

Let $X_1, ..., X_n$ be a random sample from a normal distribution with mean μ and variance σ^2. Then

- S^2 and $\bar{X}$ are independent.
- $\frac{(n-1)S^2}{\sigma^2} = \frac{1}{\sigma^2}\sum_{i=1}^{n}\left(X - \bar{X}\right)^2$ has a chi square distribution with $n - 1$ degrees of freedom.

Student t Distribution

Let Z be a standard normal random variable and let χ_n^2 be a chi square random variable with n degrees of freedom, independent of Z. The **Student t ratio with n degrees of freedom** is denoted t_n, where

$$t_n = \frac{Z}{\sqrt{\frac{\chi_n^2}{n}}}$$

is distributed as a Student t random variable with n degrees of freedom.

Let $X_1, ..., X_n$ be a random sample from a normal distribution with mean μ and standard deviation σ. Then

$$t_{n-1} = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has a Student t distribution with $n - 1$ degrees of freedom.

F Distribution

Let $\chi^2_{n_1}$ and $\chi^2_{n_2}$ be two independent chi square random variables with n_1 and n_2 degrees of freedom, respectively. A random variable of the form $\frac{\chi^2_{n_1}}{\chi^2_{n_2}}$ is said to have an **F distribution** with n_1 and n_2 degrees of freedom.

This is a right-skewed distribution with applications in ANOVA and similar tests.

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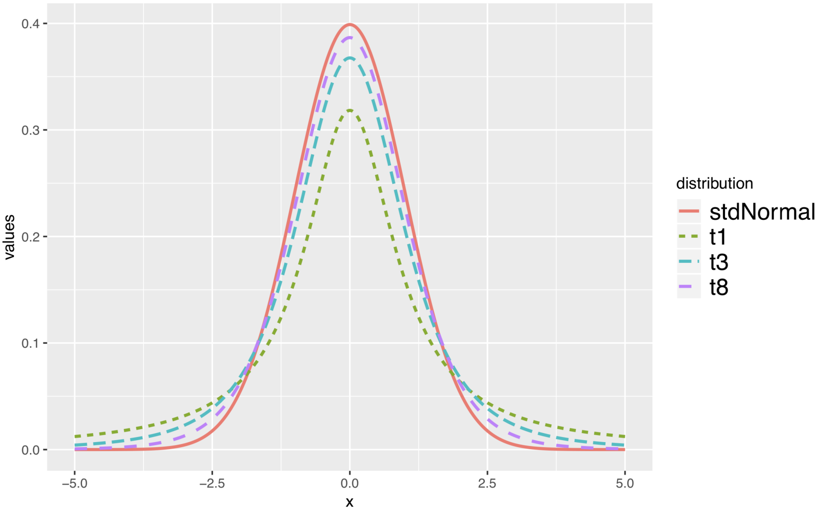
t-distribution, description

t-distribution

The t-distribution is a (standard) normal-looking probability density function that is always centered at zero, has thicker tails than the normal distribution, and has a single parameter, **degrees of freedom**.

t, plot


0.4
0.3
0.2
0.1
0.0
values
−5.0
−2.5
0.0
2.5
5.0
x
distribution
stdNormal
t1
t3
t8

t, degrees of freedom

degrees of freedom

The degrees of freedom describe the thickness of the tails of the t-distribution. The larger the degrees of freedom, the more closely the distribution looks like a standard normal distribution, $N(0, 1)$.

t, df for sample mean

If the sample has n observations and we are examining a single mean, then we use the t-distribution with $df = n - 1$ degrees of freedom.

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NOTE: Why $n - 1$? Because if you have a specified mean for a set of n numbers, you have the freedom to choose all numbers in that set **except** the last one, which can only be one possible number that depends on the others chosen. Therefore, you have $n - 1$ degrees of freedom.

For example, if your sample mean for a set of 5 numbers is $\bar{X} = 20$, you can choose whatever you like for the first four numbers, but the fifth must be whatever one number makes the sample mean 20. So if you choose 15, 25, 22, then 20, the last number **must** be 18.

t, notes

Notes on the t-distribution

- watch for skew
- thick tails of t account for estimation of σ
- must standardize random variable of interest (subtract mean, divide by sample error)

t, in R

The functions in R to deal with the t-distribution follow the same pattern as other distributions.

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?pt
?qt
?rt
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t, confidence intervals

Confidence intervals under the t-distribution are nearly identical,

$$\bar{X} \pm t^*_{1-\alpha/2,df} \cdot s_{\bar{X}}.$$

t, hypothesis tests

Hypothesis tests under the t-distribution are nearly identical, standardizing the sample statistic as before:

$$t_{df} = \frac{\bar{X} - \mu_0}{s_{\bar{X}}}.$$

t-distribution, take away

- When σ is unknown, replace Z (standard normal distribution) with t-distribution.
  - watch for skew.
- single mean degrees of freedom, $df = n - 1$
- large degrees of freedom makes t-distribution look like standard normal
- must standardize random variables with t
- safe bet: use t-distribution instead of Z everywhere!
- BUT: If you do have the option, using σ instead of s will give stronger results

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Adapted from notes by Dr. Roualdes and Dr. Gray.